

# AI-Based College Campus Lost & Found System

## Software Requirements Specification (SRS) & Module Architecture

### 1. System Architecture Overview

The system is designed using a 3-tier microservices architecture ensuring high scalability, modularity, and real-time responsiveness:

| Tier / Module    | Technology Stack                                           | Port  | Primary Responsibilities                                                     |
|------------------|------------------------------------------------------------|-------|------------------------------------------------------------------------------|
| Frontend UI      | React 19, Vite 8, React Router 7, Axios, Glassmorphism CSS | 5173  | Single Page App UI, student portal, admin dashboard & forms                  |
| Backend REST API | Node.js, Express.js, Mongoose 8, JWT, Bcrypt.js, CORS      | 5000  | Authentication, business logic, DB storage, inter-service API calls          |
| AI Microservice  | Python FastAPI, Sentence Transformers (all-MiniLM-L6-v2)   | 8000  | 384-D text embeddings, cosine similarity & keyword boosting                  |
| Database         | MongoDB Server Community Edition (v6.0+)                   | 27017 | Persistent storage for Users, Lost Items, Found Items, Claims, Notifications |

### 2. Module Specifications & Functional Requirements

#### FR-1: Authentication & User Management Module

- **Student Registration:** Collects Name, Student Register ID, College Email, Phone Number, Department, and Academic Year.
- **Password Security:** 10-round salted Bcrypt password hashing prior to persistence.
- **Session Authorization:** JWT token issued on login and attached via Bearer headers for protected endpoints.
- **Role-Based Access:** Segregates Student privileges from Administrative Verification Staff.

#### FR-2: Lost & Found Item Reporting Module

- **Lost Item Registration:** Captures item name, category, location, room, date/time, time range, brand, color, unique marks, special features, damage, and private description.
- **Found Item Logging:** Enables campus members to log found items while preserving privacy.

#### FR-3: AI Semantic Matching Engine Module

- **NLP Embeddings:** Converts item descriptions into 384-dimensional dense vectors using *all-MiniLM-L6-v2*.
- **Cosine Similarity Calculation:** Evaluates semantic distance between lost and candidate found items.
- **Keyword Boosting & Scoring:** Outputs a 0–100% Match Score and categorizes as High Potential, Possible, or Low Similarity.

#### FR-4: Privacy Protection & Redaction Module

- **Attribute Redaction:** Automatically hides sensitive details (brand, color, unique mark, damage, image URL) from non-owners to eliminate fraudulent claim attempts.

#### FR-5: Ownership Verification & Claim Review Module

- **Verification Questionnaire:** Requires claimants to answer specific hidden item parameters.
- **Weighted Scoring:** Calculates a match confidence score based on submitted answers vs. hidden found properties.

- **Admin Verification Flow:** Places claims in *pending* status for manual review and physical handover confirmation.

### 3. System Prerequisites & Hardware Specs

| Resource           | Minimum Requirement                     | Recommended Requirement                        |
|--------------------|-----------------------------------------|------------------------------------------------|
| Processor          | Intel Core i3 / AMD Ryzen 3 (Quad-core) | Intel Core i5 / AMD Ryzen 5 or higher          |
| RAM Memory         | 4 GB RAM                                | 8 GB RAM (for concurrent Node, Python & Mongo) |
| Disk Storage       | 2 GB free space                         | 5 GB free SSD space                            |
| Node.js Runtime    | v18.0.0 or higher                       | v20.x LTS                                      |
| Python Environment | v3.10.0 or higher                       | v3.12.x                                        |
| Database Engine    | MongoDB Server v6.0+ (Port 27017)       | MongoDB Atlas / Local v7.0+                    |